$$\omega_n = \frac{1}{\sqrt{LC}} = 2533.5 \text{ rad/s} \tag{1}$$

$$\zeta = \frac{r_L}{2 * \sqrt{\frac{L}{C}}} = 0.0104 \tag{2}$$

$$\omega_d = \omega_n \sqrt{1-\zeta^2} = 2533.3 \text{ rad/s} \tag{3}$$

$$Q_{factor,seriesRLC} = \frac{\omega_n L}{R} = 48.136 \tag{4}$$

$$Q_{3ph} = 3 * V_{ph} * (|I_{ph}|)^2 * X_{ph} = -602.04\text{kVAr} \tag{5}$$

$$\lambda = \{-26.316 \pm 2533\}\, \tau = \frac{L}{r_L} = 0.038 \text{ s} \tag{6}$$

KVL: $$V_{\text{dc}} \;=\; r_L\, i_L \;+\; L\, \dot{i}_L \;+\; v_C \tag{7}$$

$\Rightarrow$ $$\boxed{\dot{i}_L \;=\; \frac{1}{L}\big(-\, r_L i_L \;-\; v_C \;+\; V_{\text{dc}}\big)} \tag{1}$$

Capacitor: $$i_L \;=\; C\, \dot{v}_C \quad\Rightarrow\quad \boxed{\dot{v}_C \;=\; \frac{1}{C}\, i_L} \tag{2}$$

$$\ddot{i}_L \;=\; \frac{1}{L}\big(-\, r_L \dot{i}_L - \dot{v}_C\big) \;=\; -\frac{r_L}{L}\, \dot{i}_L \;-\; \frac{1}{LC}\, i_L$$

$$\boxed{\ddot{i}_L \;+\; \frac{r_L}{L}\, \dot{i}_L \;+\; \frac{1}{LC}\, i_L \;=\; 0}$$

Characteristic polynomial and comparison with the canonical 2nd-order form:

$$s^2 + \frac{r_L}{L} s + \frac{1}{LC} = 0 \qquad \Longleftrightarrow \qquad s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\boxed{\omega_n = \frac{1}{\sqrt{LC}}, \qquad \zeta = \frac{r_L}{2}\sqrt{\frac{C}{L}}}$$